

EX. NO: 3

POLYALPHABETIC CIPHER (VIGENERE CIPHER)

Aim:

To write a C program to implement the Vigenere Cipher, a polyalphabetic substitution technique that uses a repeating keyword to encrypt the plaintext.

Algorithm:

1. Start the program.
2. Read the plaintext and the keyword from the user.
3. Determine the length of the plaintext and the length of the keyword.
4. For every character in the plaintext, convert both the plaintext character and the corresponding keyword character (repeated cyclically) to uppercase.
5. Compute the cipher character using the relation (plaintext value + key value) mod 26.
6. Display the resulting encrypted text.
7. Stop the program.

Program:

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

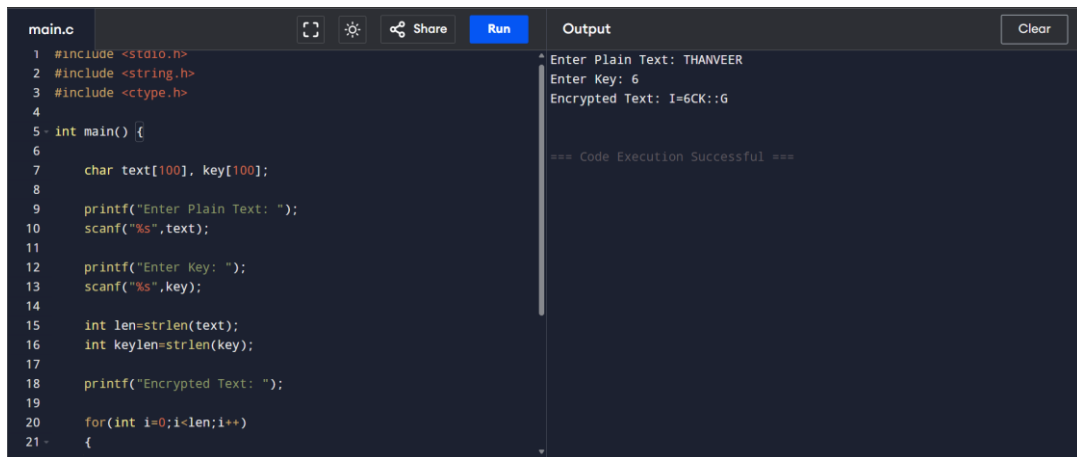
int main() {
    char text[100], key[100];

    printf("Enter Plain Text: ");
    scanf("%s", text);
    printf("Enter Key: ");
    scanf("%s", key);

    int len = strlen(text);
    int keylen = strlen(key);

    printf("Encrypted Text: ");
    for (int i = 0; i < len; i++) {
        char ch = toupper(text[i]);
        char k = toupper(key[i % keylen]);
        char enc = ((ch - 'A') + (k - 'A')) % 26 + 'A';
        printf("%c", enc);
    }
    printf("\n");
    return 0;
}
```

Output:



```
main.c  Run  Clear
1 #include <stdio.h>
2 #include <string.h>
3 #include <ctype.h>
4
5 int main() {
6
7     char text[100], key[100];
8
9     printf("Enter Plain Text: ");
10    scanf("%s",text);
11
12    printf("Enter Key: ");
13    scanf("%s",key);
14
15    int len=strlen(text);
16    int keylen=strlen(key);
17
18    printf("Encrypted Text: ");
19
20    for(int i=0;i<len;i++)
21    {
```

Enter Plain Text: THANVEER
Enter Key: 6
Encrypted Text: I=6CK::G

=== Code Execution Successful ===

Result:

Thus a C program to implement the Vigenere Cipher, a polyalphabetic substitution technique, was written, executed and the output was verified successfully.